

Distinguishing Human-Written and AI-Generated Text: A Comprehensive Study Using Explainable Artificial Intelligence in Text Classification

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Abstract—Enhancing interpretability without compromising accuracy is a critical challenge in text classification. This research explores the integration of Explainable Artificial Intelligence (XAI) techniques with advanced machine learning models, utilizing the Local Interpretable Model-Agnostic Explanations (LIME) framework to provide transparency. A fine-tuned BERT model achieved state-of-the-art performance, surpassing Random Forest and Sentence Embedding-based models with a perfect 100% accuracy (ROC-AUC score of 1.00). While Random Forest classifiers offered a solid baseline, they struggled with semantic nuances, underscoring the need for embedding-based approaches. The study highlights the inherent trade-off between interpretability and accuracy, demonstrating that while transformer-based models like BERT excel at capturing complex linguistic patterns, their "black-box" nature necessitates tools like LIME for explainability. By bridging this gap, the research contributes to the development of more transparent, reliable, and high-performing AI systems.

Index Terms—Explainable AI, Human vs AI Text Classification, LIME, Random Forest Classifier, BERT, Machine Learning Interpretability, Semantic Text Embeddings

I. INTRODUCTION

The growing reliance on machine learning models for decision-making has created a demand for systems that are not only accurate but also interpretable and transparent. In critical fields like healthcare, finance, and law, understanding why a model makes a prediction is crucial. However, powerful models like Random Forests and BERT often lack transparency. Explainable AI (XAI) tackles this by making model outputs more understandable. Techniques like LIME help explain individual predictions without altering the original model.

Transformer-based architectures, particularly BERT, have revolutionized natural language processing (NLP) by delivering state-of-the-art performance in numerous applications. However, these models pose challenges in explainability due to their complex attention mechanisms and high-dimensional representations. In contrast, Random Forests offer a balance between interpretability and accuracy but still benefit from the additional explanatory capabilities provided by LIME.

This study employs LIME to interpret the predictions of Random Forest classifiers and fine-tuned BERT models, both of which have shown strong performance on a specific text classification task. By combining high accuracy with enhanced

explainability, this work contributes to the advancement of interpretable and reliable machine learning systems.

II. LITERATURE REVIEW

The rapid evolution of Explainable Artificial Intelligence (XAI) reflects the increasing demand to balance the predictive power of machine learning models with the transparency required for trust and accountability. Among XAI techniques, Local Interpretable Model-Agnostic Explanations (LIME) has gained significant attention for its ability to provide local explanations for complex algorithms. By perturbing input data and training interpretable surrogate models, LIME has demonstrated its utility across various domains, including healthcare, finance, and natural language processing [1].

Random Forest classifiers, introduced by Breiman [2], are ensemble methods that combine multiple decision trees to produce robust predictions. Their simplicity and ability to handle noisy data make them a popular choice for classification tasks [3]. However, the interpretability of Random Forest models is often limited due to their ensemble nature, necessitating the use of supplementary tools like LIME [4]. Studies have shown that integrating Random Forests with LIME enhances their interpretability by providing insights into feature importance [5], [6].

Transformer-based architectures, particularly BERT, have revolutionized natural language processing tasks, achieving state-of-the-art performance across benchmarks such as text classification and sentiment analysis [7], [8]. Despite their accuracy, the deep architecture of transformers makes them challenging to interpret [9]. XAI techniques such as LIME and SHAP have been used to explain BERT's predictions, improving their applicability in critical areas [10], [11].

LIME has been widely compared to other interpretability techniques. Lundberg and Lee [11] introduced SHAP, a method rooted in cooperative game theory that provides both global and local explanations. However, LIME's simplicity and versatility have made it a more accessible choice for a broader range of models [12]. Comprehensive reviews by Guidotti et al. [13] and Gilpin et al. [14] highlight LIME's effectiveness in providing quick, localized explanations for real-world applications.

The integration of LIME with hybrid machine-learning models has yielded promising results. Satish and Shafqaat [15] demonstrated BERT-based predictions for fake news detection, while Feyza et al. [16] successfully combined LIME and SHAP with Random Forests, Gradient Boosting for medical diagnostics, offering actionable insights to healthcare professionals.

The ethical implications of AI systems have further emphasized the importance of interpretability and accountability. Doshi-Velez and Kim [6] stressed the role of transparency in fostering trust, particularly in high-stakes applications such as healthcare and autonomous vehicles. LIME’s ability to provide intuitive, instance-level explanations directly addresses these concerns.

Advancements in XAI have also sought to enhance the robustness and efficiency of LIME. Ribeiro et al. [17] extended LIME to handle temporal data, while Alvarez-Melis and Jaakkola [18] explored causal interpretability techniques. These developments expand LIME’s applicability, making it a versatile tool in modern AI systems.

Building on this foundation, the present study leverages LIME to interpret the predictions of Random Forest and fine-tuned BERT models for text classification tasks. By integrating high accuracy with enhanced interpretability, this research contributes to the ongoing pursuit of responsible and transparent AI systems.

III. METHODOLOGY

A. Dataset Description:

This study utilizes the LLM Detect AI-Generated Text dataset [19], a benchmark corpus curated to distinguish machine-generated text from human-authored content. The dataset contains 44,868 documents in total, where human-written text (label 0) accounts for 61.0004% (27,371 samples), and AI-generated text (label 1) constitutes 38.9996% (17,497 samples).

Further categorization includes *Persuade_corpus*, which represents 58% of the data, *Mistral7bInstruct_v2* with 5%, and other sources making up 37%. These attributes highlight the diversity of the dataset, essential for distinguishing between human and AI-generated text effectively.

TABLE I
DETAILS OF DATASET FEATURES.

Feature	Value
Text	essay text
Label	1 for AI generated, 0 for human
Prompt Name	original persuade prompt
Source	source dataset

B. Data Preprocessing:

The dataset $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ consists of $N = 44,858$ text samples, where each sample is labeled either as human-authored ($y = 0$) or AI-generated ($y = 1$). To ensure a balanced representation of both classes in the training and testing sets, stratified splitting was applied. This method preserves the

original class proportions during the split. The process resulted in the creation of two subsets, $\mathcal{D}_{\text{train}}$ and $\mathcal{D}_{\text{test}}$, with 30% of the dataset allocated for testing. The parameter `stratify=y` was specifically used to maintain the class ratio in both subsets.

In the feature extraction phase, TF-IDF (Term Frequency-Inverse Document Frequency) vectorization was employed with bi-grams to represent the text samples numerically. The TF-IDF score measures the importance of a term within a document relative to the entire dataset. Mathematically, it is computed using the equation:

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \times \log \left(\frac{N}{\text{DF}(t) + 1} \right) \quad (1)$$

Here, $\text{TF}(t, d)$ refers to the frequency of the term t in the document d , while $\text{DF}(t)$ denotes the number of documents containing the term t . The term N represents the total number of documents in the dataset. The logarithmic scaling reduces the weight of terms that are commonly found across multiple documents, ensuring that less frequent but more relevant terms receive greater emphasis in the representation.

For feature extraction using BERT (Bidirectional Encoder Representations from Transformers), the text samples were tokenized using the WordPiece algorithm. Each text sequence was tokenized with a maximum length of $l = 512$ tokens. The tokenized representation includes special tokens such as [CLS], which is added at the beginning of the sequence to represent the overall input for classification tasks, and [SEP], which indicates the end of the sequence or separates sentences. The mathematical representation of the BERT tokenization process can be written as:

$$\mathbf{E}_{\text{BERT}} = \text{Tokenizer}(x_i) \rightarrow \{[\text{CLS}], w_1, \dots, w_l, [\text{SEP}]\} \quad (2)$$

In this representation, x_i refers to an individual text sample from the dataset, and $w_1, \dots, w_l$ are the tokenized subwords, where l is the maximum sequence length. The inclusion of special tokens and the subword tokenization ensures that the text is appropriately processed for input into the BERT model. These features extracted from both TF-IDF and BERT tokenization were used for further processing and analysis in the study.

C. Random Forest Classifier:

The Random Forest classifier used in this study consists of an ensemble of $T = 200$ decision trees, trained to minimize Gini impurity at each node. Gini impurity is calculated using the equation:

$$\text{Gini}(t) = 1 - \sum_{c=0}^1 (p(c|t))^2 \quad (3)$$

In this equation, $p(c|t)$ represents the probability of class c at node t . This measure evaluates the homogeneity of the samples at a given node, with lower values indicating purer splits.

The training process incorporates several hyperparameters. The maximum depth of the trees is set to be unlimited,

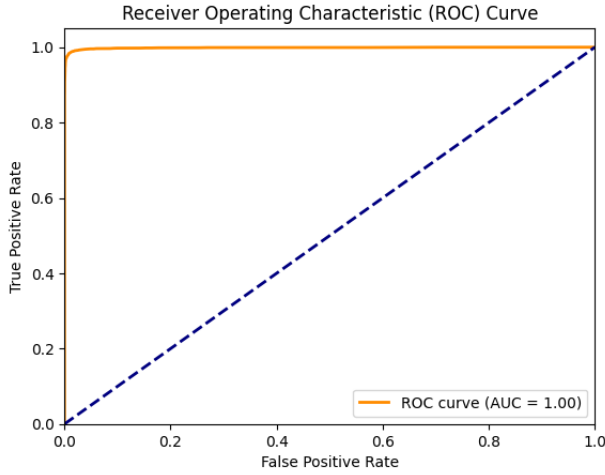


Fig. 1. ROC Curve of the Random Forest Classifier.

allowing full tree expansion. Feature sampling is defined as $\sqrt{d}$ features per split, where d corresponds to the dimensionality of the TF-IDF feature space. Additionally, class weights are balanced to address the minority class ($y = 1$), ensuring the classifier does not favor the majority class.

The final prediction is made by aggregating the votes from all decision trees in the ensemble, following the majority rule:

$$\hat{y} = \text{mode}(\{h_t(x)\}_{t=1}^T) \quad (4)$$

Here, $h_t(x)$ denotes the prediction from the t -th decision tree, and the mode function selects the most frequent class label among the predictions. This approach enhances the robustness and accuracy of the classifier by leveraging the collective decision-making of multiple trees. Here figure 1 shows ROC Curve of the Random Forest Classifier.

D. Embedding (all-MiniLM-L6-v2) with Random Forest Classifier:

Let $\mathcal{X}$ denote the text input space and $\mathcal{Y} = \{0, 1\}$ represent the classification labels. The combined MiniLM-RF model is defined as a composite function:

$$f_{\text{MiniLM-RF}} : \mathcal{X} \xrightarrow{\Phi} \mathbb{R}^{384} \xrightarrow{\Psi} \mathcal{Y} \quad (5)$$

This function consists of two main components: the embedding transformation Φ and the Random Forest classifier Ψ .

1) *Embedding Transformation Φ :* The MiniLM-L6-v2 model Φ maps an input text $x \in \mathcal{X}$ to a dense semantic vector representation in a 384-dimensional space. This transformation is expressed as:

$$\Phi(x) = \text{LayerNorm}(\mathbf{W}_p \cdot \text{Transformer}^6(x)_{[\text{CLS}]}) \quad (6)$$

In this equation, Transformer^6 refers to the 6-layer distilled transformer network that processes the input text and outputs contextual embeddings, $\mathbf{W}_p \in \mathbb{R}^{384 \times d_h}$ is the projection matrix that reduces the hidden dimension d_h of the transformer output to 384 dimensions, LayerNorm is the layer normalization applied to stabilize and normalize the output embeddings

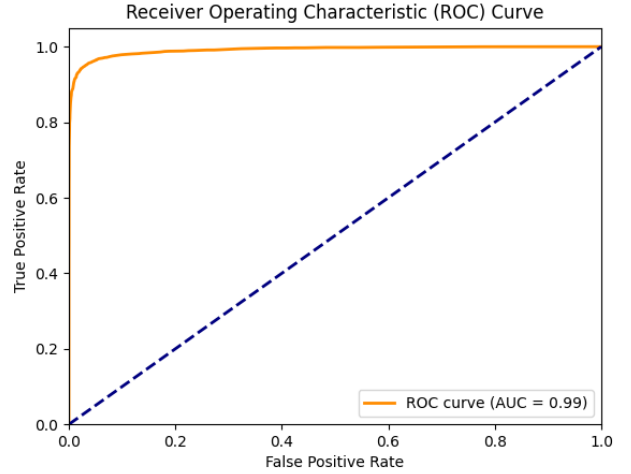


Fig. 2. ROC Curve for Embedding (all-MiniLM-L6-v2) with Random Forest Classifier.

and [CLS] represents the special classification token, which is used to aggregate information about the entire input sequence.

2) *Random Forest Classifier Ψ :* The Random Forest (RF) classifier operates on the dense embeddings in the space $\mathbb{R}^{384}$. It predicts the classification labels by aggregating the outputs of $T = 200$ decision trees. The prediction is determined by majority voting:

$$\Psi(\mathbf{e}) = \text{mode}(\{h_t(\mathbf{e})\}_{t=1}^{200}) \quad (7)$$

Each decision tree h_t partitions the embedding space through axis-aligned splits by minimizing the Gini impurity at each node:

$$\text{Gini}(t) = 1 - \sum_{c=0}^1 \left(\frac{N_c^t}{N^t} \right)^2 \quad (8)$$

In this expression, N_c^t denotes the number of samples of class c at node t and N^t represents the total number of samples at node t . The hyperparameters of the RF classifier include the total number of trees $T = 200$, the number of features per split $m = \lfloor \sqrt{384} \rfloor = 19$, and unbounded tree depth, allowing full expansion. The split criterion at each node is determined by selecting the feature and threshold that minimize the Gini impurity, ensuring optimal partitions in the embedding space. Figure 2 shows ROC curve for embedding (all-MiniLM-L6-v2) with RF classifier.

E. Embedding (BERT) with Random Forest Classifier:

The BERT-RF framework integrates contextual embeddings generated by BERT with ensemble classification using a Random Forest classifier. The overall model is represented as a composite function:

$$f_{\text{BERT-RF}} : \mathcal{X} \xrightarrow{\Phi_{\text{BERT}}} \mathbb{R}^{768} \xrightarrow{\Psi_{\text{RF}}} \mathcal{Y} \quad (9)$$

1) *BERT Embedding Process* Φ_{BERT} : For an input text $x = [w_1, \dots, w_n]$, where w_i represents the words in the sequence, the BERT model generates contextual embeddings. These embeddings are processed as follows:

$$\Phi_{BERT}(x) = \mathbf{h}_{[CLS]} = \text{LayerNorm}(\mathbf{W}_p \cdot \text{Transformer}^{12}(x)_{[CLS]}) \quad (10)$$

In this formulation, Transformer^{12} is the 12-layer bidirectional transformer network in the BERT architecture, which encodes the input sequence into contextual embeddings, $\mathbf{h}_{[CLS]} \in \mathbb{R}^{768}$ represents the pooled output of the special classification token [CLS], summarizing the input sequence for classification tasks, $\mathbf{W}_p \in \mathbb{R}^{768 \times d_h}$ is a projection matrix that maps the hidden dimension d_h of the transformer output to the 768-dimensional embedding space and LayerNorm is the layer normalization applied to stabilize and normalize the embedding output.

2) *Random Forest Classification* Ψ_{RF} : The Random Forest classifier operates on the 768-dimensional embeddings produced by BERT. The prediction is based on majority voting across $T = 200$ decision trees:

$$\Psi_{RF}(\mathbf{h}) = \text{mode}(\{h_t(\mathbf{h})\}_{t=1}^{200}) \quad (11)$$

Each decision tree h_t partitions the embedding space by optimizing the Gini impurity for left (s_L) and right (s_R) child nodes:

$$s^* = \arg \min_{s \in S} [\text{Gini}(s_L) + \text{Gini}(s_R)] \quad (12)$$

The Gini impurity for a node s is calculated as:

$$\text{Gini}(s) = 1 - \sum_{c=0}^1 (p_c^{(s)})^2, \quad p_c^{(s)} = \frac{N_c^{(s)}}{N^{(s)}} \quad (13)$$

Here, $N_c^{(s)}$ is the count of samples belonging to class c at node s , and $N^{(s)}$ is the total number of samples at the node. The probability $p_c^{(s)}$ reflects the class distribution within the node.

Key parameters of the Random Forest implementation include a total of 200 trees, 27 features per split (calculated as $\lfloor \sqrt{768} \rfloor$), and unconstrained tree depth, allowing full growth. The optimized splits at each node ensure the classifier effectively separates the embeddings based on class labels. Figure 3 shows embedding (BERT) with RF classifier.

F. BERT Fine-Tuning:

The BERT-base architecture, consisting of 110 million parameters, 12 layers, and 768-dimensional embeddings, was fine-tuned using a supervised learning approach with cross-entropy loss. The loss function is defined as:

$$\mathcal{L} = -\frac{1}{B} \sum_{i=1}^B \sum_{c=0}^1 y_{i,c} \log(\sigma(\mathbf{W}_c \mathbf{h}_{[CLS]} + b_c)) \quad (14)$$

In this equation, $\mathbf{h}_{[CLS]} \in \mathbb{R}^{768}$ represents the pooled embedding from the special classification token [CLS], which summarizes the input sequence, $\mathbf{W} \in \mathbb{R}^{2 \times 768}$ is the classification head, a weight matrix mapping the 768-dimensional embeddings to two output classes, b_c denotes the bias term for

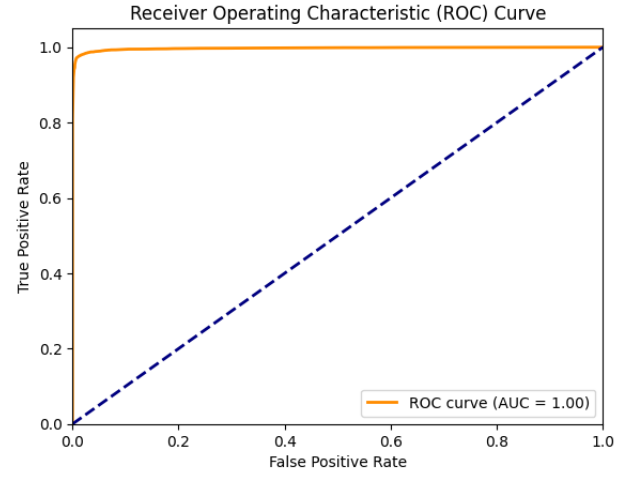


Fig. 3. ROC Curve for Embedding (BERT) with Random Forest Classifier.

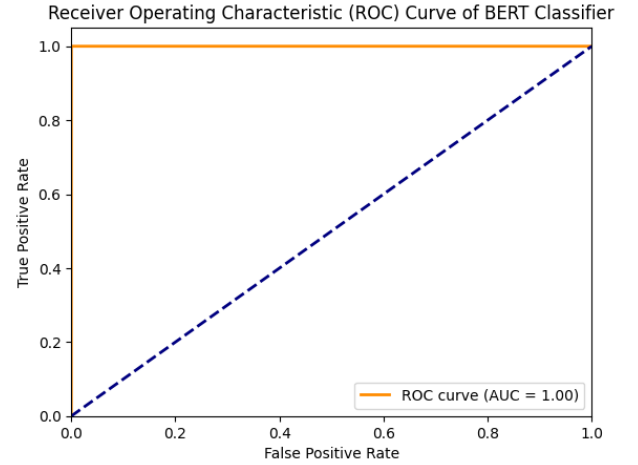


Fig. 4. ROC Curve for the Fine-Tuned BERT Model.

class c , σ is the softmax activation function, which converts the logits into class probabilities, $y_{i,c}$ is the ground-truth label for the i -th sample and class c and B is the batch size, set to 16 in this implementation.

The fine-tuning process utilized the AdamW optimizer, which is well-suited for training large transformer models. The optimizer was configured with hyperparameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$ to ensure stable and efficient updates to the model's weights. Additionally, a linear learning rate warmup strategy was applied, gradually increasing the learning rate during the initial training steps for better convergence. The model was fine-tuned for three epochs to optimize its performance on the classification task. Figure 4 shows ROC curve for the fine-tuned BERT model.

G. Model Interpretability with LIME:

The Local Interpretable Model-agnostic Explanations (LIME) technique was applied to enhance the interpretability of the classifier $f : \mathcal{X} \rightarrow \mathcal{Y}$. LIME generates local

explanations by approximating the model’s behavior in the vicinity of a given input x using a simpler, interpretable model $g \in \mathcal{G}$. This is achieved by solving the following optimization problem:

$$\xi(x) = \arg \min_{g \in \mathcal{G}} \underbrace{\mathbb{E}_{z \sim \pi_x} [\mathcal{L}(f(z), g(z))]}_{\text{Fidelity}} + \underbrace{\Omega(g)}_{\text{Complexity}} \quad (15)$$

In this formulation, π_x represents a Gaussian kernel that defines the locality around the input x , assigning higher weights to perturbed samples closer to x , $\mathcal{G}$ is the class of interpretable models, typically linear models, used to approximate the classifier’s decision boundary in the local region, $\mathcal{L}$ is the loss function, which measures the difference between the predictions of the original model f and the simpler model g . The term $\Omega(g)$ penalizes the complexity of the simpler model g , ensuring that the explanation remains interpretable and $\xi(x)$ denotes the final explanation generated by LIME for the input x . For this implementation, $K = 10$ top features were selected to construct the explanation, providing insight into the most influential factors driving the model’s predictions. Additionally, $S = 5000$ perturbed samples were generated for each input to estimate the local decision boundary effectively.

H. Evaluation Protocol:

The performance of the model was evaluated using multiple metrics to provide a comprehensive assessment of its predictive capabilities. First, the confusion matrix was used to quantify the key classification outcomes: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). Second, Receiver Operating Characteristic (ROC) analysis was conducted to evaluate the model’s discrimination ability across different thresholds. The Area Under the Curve (AUC) was computed using the trapezoidal rule, as given by:

$$\text{AUC} = \int_0^1 \text{TPR}(f) \cdot \text{FPR}'(f) df \quad (16)$$

Here, TPR denotes the True Positive Rate, and FPR denotes the False Positive Rate, which together characterize the trade-off between sensitivity and specificity. Finally, classification metrics, including precision, recall, and the F1-score, were calculated to assess the model’s accuracy, sensitivity, and balance between precision and recall

IV. EXPERIMENTAL RESULTS AND ANALYSIS

The results demonstrate that the fine-tuned BERT model significantly outperforms both the Random Forest and Sentence Embedding-based models across all evaluation metrics. The BERT model achieves an accuracy of 100% (ROC, AUC score of 1.00), underscoring its superior ability to distinguish between classes with remarkable precision. In comparison, the Random Forest model, while effective when utilizing TF-IDF features, struggled to capture the deeper semantic relationships present in the data, limiting its overall performance. Sentence Embedding-based models achieved competitive results, emphasizing the critical role of semantic representations in

enhancing text classification tasks. Figure 7 shows the accuracy, precision, recall, and F1 score comparison among various models.

An important observation from the analysis is the inherent trade-off between model interpretability and performance. Random Forest models, with their reliance on TF-IDF features, provide better transparency by offering clear insights into feature importance. However, transformer-based models like BERT prioritize predictive accuracy, often at the cost of reduced interpretability. This trade-off highlights the necessity of choosing models based on the specific requirements of the task, balancing the need for accuracy with the demand for explainability. Figure 5, 6 show the LIME Output of the explanation of an AI-Generated Text. Figure 8 shows an example of the significance of weight in LIME explanations.

TABLE II
CLASS-WISE PERFORMANCE METRICS FOR RANDOM FOREST CLASSIFIER AFTER HYPERPARAMETER TUNING.

Class	Precision	Recall	F1-score	Support
Human	0.97	1.00	0.99	8212
AI	1.00	0.96	0.98	5249
accuracy			0.98	13461
Macro avg	0.99	0.98	0.98	13461
Weighted avg	0.98	0.98	0.98	13461

TABLE III
CLASS-WISE PERFORMANCE METRICS FOR BERT EMBEDDINGS WITH RANDOM FOREST CLASSIFIER.

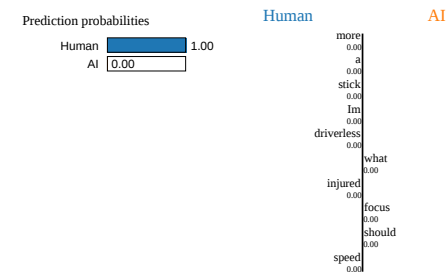
Class	Precision	Recall	F1-score	Support
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accuracy			0.98	13461
Macro avg	0.98	0.98	0.98	13461
Weighted avg	0.98	0.98	0.98	13461

TABLE IV
CLASS-WISE PERFORMANCE METRICS FOR FINE-TUNED BERT MODEL.

Class	Precision	Recall	F1-score	Support
Human	1.00	1.00	1.00	8270
AI	1.00	1.00	1.00	5191
accuracy			1.00	13461
Macro avg	1.00	1.00	1.00	13461
Weighted avg	1.00	1.00	1.00	13461

V. CONCLUSION

This study highlights the effectiveness of fine-tuned BERT in distinguishing between AI-generated and human-authored text, achieving superior performance across all evaluation metrics. Semantic-rich embeddings from Sentence Transformers also contributed positively to classification tasks. Transformer-based models like fine-tuned BERT excel in handling complex language patterns, while ensemble methods like Random Forest, though effective as baselines, show limitations with deeper semantic features. Embedding-based approaches demonstrate a balance between interpretability and accuracy, making them valuable for text classification tasks.



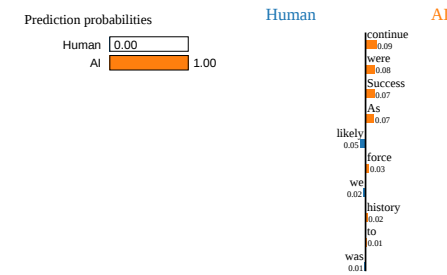
Text with highlighted words

Driverless cars

You may think that **driverless** cars are good but I think that **driverless** cars are senseless I wouldnt want that I would want to drive my car myself. I waited **a** long time to get my driving license and drive my own car but no they had to come up with the **driverless** car and make all the other cars seem useless. **im** going to figure out **what** makes this car so good, and see if it would come in handy because its alot of people are craving this car.

This car seem so crazy because **what** if my car starts to act all crazy and catch on fire **what** am i going to do im waisted my money and time on that car and it would be hard to buy **a** new one. I think we should all this **stick** with the cars we are use to. Because when you get this car you wont know **what** to do, and most people dont read manual to cars they just say they figure it out on their own. Once they try that they end up messing something up and having to spend **more** money on getting it fixed.

Fig. 5. LIME Output Illustrating the Explanation of a Human-Generated Text.



Text with highlighted words

Winston Churchill, one of the greatest leaders in **history**, famously said, "**Success** is not final, failure is not fatal: it is the courage to **continue** that counts." This statement highlights the importance of failure in the pursuit of success. Failure is a natural part of the journey to success, and it is how **we** respond to it that matters.

An individual's enthusiasm level plays a crucial role in their ability to succeed. Enthusiasm is the driving **force** behind any successful endeavor. It is the passion and energy that propels us to keep going even when things get tough. When **we** are enthusiastic about something, **we** are more **likely** to put in the effort and persist through setbacks.

For example, when I started my own business, I was incredibly enthusiastic about the idea. I had a clear vision of what I wanted to achieve, and I was determined to make it happen. However, there **were** many times when I faced obstacles and setbacks. I could have given up, but my enthusiasm kept me going. I believed in my idea, and I was willing to put in the hard work to make it a reality.

Fig. 6. LIME Output Illustrating the Explanation of an AI-Generated Text.

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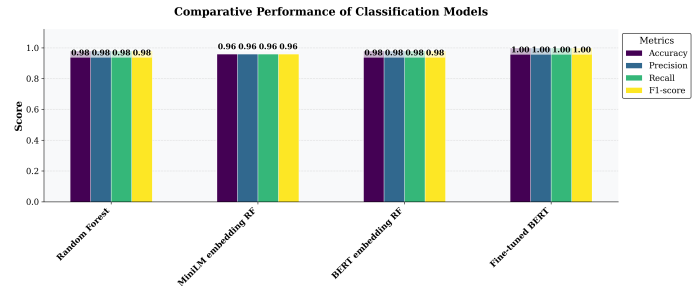


Fig. 7. Comparison of Model Performance.

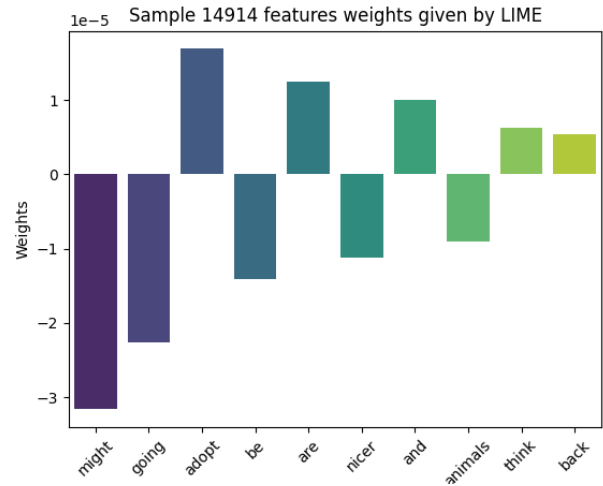


Fig. 8. Significance of Feature Weights in LIME Explanations.

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